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Research Topic: CRISPR-Cas12a-Based Detection of Gilbert Syndrome-Associated UGT1A1 Variants: Development and Preliminary Validation of a Rapid Molecular Screening Platform

CRISPR-Cas12a-Based Detection of Gilbert Syndrome-Associated UGT1A1 Variants: Development and Preliminary Validation of a Rapid Molecular Screening Platform

Introduction

Bilirubin conjugation disorders are common, non-pathogenic disorders, called Gilbert syndrome (GS), in which variants in the promoter of UGT1A1 with lower function are present, the most common one being the TA-repeat expansion (UGT1A1*28). UGT1A1 genotype is known to be a pharmacogenomic predictor of the toxicity of irinotecan and atazanavir, and a hepatic phenotype that does not cause toxicity. The current methods of confirmatory genotyping are mainly based on capillary electrophoresis, Sanger sequencing, and real-time PCR or high-resolution melting. All of which are centralised in molecular laboratories. We report the design, assembly, and initial validation of an isothermal CRISPR-Cas12a-based genotypic assay for the UGT1A1 TA-repeat polymorphism that can be performed rapidly and without a central lab.

Research Problem

Bilirubin is conjugated to produce bile acids in the liver by UDP-glucuronosyltransferase 1A1 (UGT1A1). There is a well-characterised polymorphism of the TATA box promoter of UGT1A1, with a two base pairs (TA) insertion compared to the wild-type with six repeats of the same sequence (UGT1A1*28); the rarer forms of the enzyme (five and eight-repeat alleles UGT1A1*36 and UGT1A1*37) are associated with reduced promoter activity and liver expression and is the major genetic basis for Gilbert syndrome, the most common inherited cause of isolated unconjugated hyperbilirubinemia [1]-[4]. The implications of this locus are much more than just “benign jaundice”. Another function of UGT1A1 is the metabolism of the active metabolite of the chemotherapeutic drug irinotecan (SN-38). Patients with alleles with less enzyme activity are very prone to severe neutropenia and diarrhoea. International pharmacogenomic guidelines for irinotecan now recommend or require UGT1A1 genotyping prior to administration of irinotecan [5,6]. The same locus is involved in the modulation of the risk of hyperbilirubinemia and treatment discontinuation in HIV-infected patients with atazanavir treatment [8]. Despite this, a number of recent reviews have consistently identified poor adherence to and uptake of guidelines, unavailability of laboratories, and reimbursement problems. Testing workflow is a significant barrier to the routine implementation of UGT1A1 genotyping in non-well-resourced health systems [6].

All confirmatory methods currently available are analytically valid, but require the use of a thermal cycler, fluorescent instrumentation, and reagents that need to be kept cold. Molecularly trained personnel normally deliver results in 1 to 2 days or more after sample collection [1, 3, 4, 9]. The turnaround and infrastructure requirements of centralised molecular diagnostics, which are available in a few tertiary centres in countries such as Bangladesh, limit the opportunity for the pre-emptive molecular diagnosis of patients who need pre-emptive diagnosis to be started on irinotecan or protease inhibitor therapy and reassure patients with incidental hyperbilirubinemia

who would otherwise be unnecessarily escalated to imaging or liver biopsy. There is a clear need for a rapid, equipment-light, and point-of-care deployable genotyping method for a clinically important locus with a clear need: the UGT1A1 TA-repeat locus.

Literature Review

The genetic basis of Gilbert syndrome has been well defined, as has its potential pharmacogenomic significance. Both the catalogue of the A(TA)7TAA UGT1A1 promoter allele(s) and its association with decreased UGT1A1 expression and hyperbilirubinemia [1,2] as well as following papers on the development of molecular assays, optimisation of throughput and turnaround: TaqMan-based real-time PCR assay for UGT1A1*28/*36/*37 [3] and HRM-based (TA)n genotyping validated against Sanger sequencing on clinical samples [4] and multi-platform, external quality assessment programmes comparing the methods for UGT1A1 testing in relation to irinotecan [9] are considered to be clinical genotyping resources. At the same time, the field of HIV has shown associations between atazanavir and UGT1A1 [8]. In the literature, there was evidence of genotype dosing in the case of irinotecan [5]-[7] and HIV; lack of access to testing was identified as the major unresolved implementation barrier [6].

Another emerging body of literature focuses on diagnosing nucleic acids using CRISPR-Cas systems. Using conventional sequencing, Chen et al. demonstrated that Cas12a can indiscriminately trans-cleave single-stranded DNA reporters, enabling the DETECTR platform to quickly and sequence-specifically detect DNA [11]. Gootenberg et al. built on this and created a multiplexed platform using Cas13 and Cas12a. Csm6, which allows for the portable and quantitative detection of nucleic acids with a lateral flow readout [12]. Since then, the platforms have been adapted for SNVs, including in oncology for detection of BRAFV600E mutations [13] and in infectious diseases for the single-nucleotide discriminatory power of the readout from a lateral-flow and smartphone-based detection platform for IDH1 mutant and wild-type alleles intraoperatively [14]. The systematic optimization of the crRNA length and valency has shed light on the architecture of the crRNA in its ability to bind to and discriminate between Cas12a [15]. A combination of isothermal recombinase polymerase amplification (RPA) and lateral-flow strips has also been field-validated in combination with CRISPR-Cas12a to detect infectious targets, such as Leptospira lipL32 [16], Burkholderia pseudomallei in a low- and middle-income country (LMIC) diagnostic benchmarking study [17]. Chili leaf curl virus in crude plant extracts [18], and this RPA-Cas12a-LFA combination is a mature and transferable diagnostic architecture. At the junction of these two literatures, it becomes apparent that there is a gap between the two. A few experiments with Cas12a-based SNVs were published with respect to short tandem repeats (STRs) found in the low-complexity promoter region, such as the TATA box of the UGT1A1 gene, but not with respect to short tandem repeats (STRs) in the coding sequence. The author is not known as yet of any published assays available that are based on engineered crRNA, Cas12a trans-cleavage detection for the UGT1A1 (TA)n repeat of Gilbert syndrome and the irinotecan/atazanavir pharmacogenomic risk. This is a translational opportunity not yet considered: transfer of a well-established CRISPR-diagnostic architecture from single base

substitution and pathogen targets to a clinically relevant repeat-length pharmacogenetic polymorphism.

Research Question

- Is it possible to distinguish the UGT1A1 promoter TA-repeat genotypes (TA6/TA6 wild type, TA6/TA7 heterozygous, and TA7/TA7 homozygous UGT1A1*28, and the less common TA5 and TA8 genotypes) using a CRISPR-Cas12a trans-cleavage assay that can be performed with isothermal recombinase polymerase amplification and an optimized crRNA without the need for thermal cycling and sequencing instrumentation?

Three sub-questions support the proposed methodology for this primary question,

- Whether the alleles are only 2-bp different at the TA binding site, and whether Cas12a's 2-bp tolerance for PAM-distal mismatches is sufficient to distinguish them, given its known capability to tolerate 2-bp differences at the PAM-distal site when using crRNAs with the engineered seed regions?
- What is the most appropriate pre-amplification chemistry to use to ensure fidelity of the AT-rich TATA-box repeat region and to minimize the appearance of polymerase-slippage artefacts that will make the determination of the repeat length more difficult?
- How is the analytical performance of the resulting assay in terms of sensitivity, specificity, turnaround time and inter-operator reproducibility of the test compared with the reference method (capillary electrophoresis or Sanger sequencing) on a pilot panel of genomic DNA samples?

Proposed Methodology

Phase one: Design of crRNA and specificity of screening

A modelling approach will be used in silico to model the UGT1A1 promoter (TA)n repeat locus (UGT1A1 5' regulatory region) at the level of the chromosome. Like ARTEMIS and SDS-CRISPR, deliberate mismatches will be designed in the seed region for the candidate crRNAs to bias the trans-cleavage activation of Cas12a in favour of a certain repeat length (e.g., TA7 is more strongly activated than TA6) [13] [14]. On-target/off-target discrimination margin will be computed using computational screening. The multiple crRNA candidates will be ranked before they are synthesized using the reference genome of the human genome.

Phase 2: The assembly of the assay and validation of the analytical method

The primers for the (TA)n repeat will be designed and optimised at 37-42℃ and isothermal alternative methods (e.g., isothermal amplification by polymerase cycling) are not considered, as there is a high risk of polymerase slippage, which will lead to a distortion of the apparent repeat length in this AT-rich low-complexity target region. The amplified product will be mixed with a selected crRNA and recombinant Cas12a (LbCas12a/AsCas12a). A single-stranded DNA reporter molecule that is complementary to the crRNA, with a quenched fluorophore such as biotin. Both a quantitative kinetic characterisation based on real-time fluorometry and a lateral-flow strip that can be read out in the field without any equipment will be used to collect

readout in parallel as before for the RPA-Cas12a-lateral-flow diagnostics [16]-[18]. The LAOD will be calculated along with the analytical specificity between repeat lengths of adjacent alleles (alleles 5, 6, 7, and 8, which are the percent of the population of each of these alleles), and the total assay turnaround time will be determined using synthetic alleles (TA) under controlled conditions.

Phase 3: Preliminary clinical validation

The partnering lab will then send a set of de-identified genomic DNA samples (approx. 30-50) from each individual to another diagnostic facility for a blind test with the constructed (TA)n assay. The results from the concordance with the reference genotype will be reported as sensitivity, specificity, positive or negative predictive value, and Cohen's kappa, and inter-day and inter-operator reproducibility will be assessed by repeat testing of a subset of samples. We will discuss turnaround time, approximate reagent costs per sample, and comparison with the reference method. All samples will be processed according to the guidelines of the institutional ethics committee and informed, anonymized, and handled in a typical fashion as is routinely practiced in genetic studies of human subjects.

Phase 4: Data Analysis

Diagnostic performance will be given at the per-genotype level in 2×2 contingency tables and will have precise (exact) confidence intervals for sensitivity and specificity based on the expected pilot sample size. In Phase Two, the data will be used to make a cutoff for the signal-to-threshold, and the data for the unblinded comparison in Phase Three will be used to infer the genotype.

Research Topic

The work proposed is novel in two aspects. **First,** it opens up a new class of CRISPR-Cas12a diagnostics, as previously demonstrated for detecting canonical single-nucleotide substitutions in oncology, to a short tandem-repeat (STR) promoter polymorphism, which is technically more challenging due to the low complexity (AT-rich) sequence context and the need for high-fidelity isothermal amplification, and the ability of Cas12a to tolerate mismatches at the PAM-distal position. **Second,** it is not a format of an infectious disease assay; rather, it is a custom workflow that was developed for a specific target (the seed-region crRNA) and an engineered assay for a repeat length pharmacogenetic locus that was read out with a lateral-flow/fluorescence format. The subject is becoming clinically manifest and relevant to the health systems. The low availability of UGT1A1 genotyping has been identified as the biggest constraint to implementing these guidelines in each implementation review [5]-[8] and has led to their use in guiding the risk of hyperbilirubinemia in relation to atazanavir use. A decentralised, same-day, equipment-light confirmatory test for the UGT1A1 TA-repeat genotype would directly tackle this access gap, especially in health systems where access to molecular genotyping is limited, and provide a quicker, less-invasive method for establishing diagnostic certainty of benign Gilbert syndrome in individuals with incidental unconjugated hyperbilirubinemia, avoiding unnecessary downstream investigation. Beyond the short repeat length polymorphisms, a way of making the Cas12a have

a more specific activity around other pharmacogenes would make the platform more specific than Gilbert syndrome.

Expected Outcomes

- A library of engineered crRNAs that can be used to discriminate between the alleles TA6 and TA7 (and ideally TA5/TA8) of UGT1A1 using the Cas12a trans-cleavage mechanism has been validated.
- Optimized RPA-Cas12a-lateral flow with clearly defined LOD, analytical specificity, and turnaround time (TAT, <60-90 min, without the need for a thermal cycler).
- Pilot concordance data (sensitivity, specificity, kappa) demonstrated the feasibility of decentralised UGT1A1 genotyping using the pilot data (samples ~30-50).

Limitations

- The statistical power and generalisability of the sample size are small, which reduces generalisability across different populations with varying allele frequencies.
- In contrast to SNV detection, engineered mismatches for repeat-length discrimination are more difficult to obtain, and there is the potential for residual cross-reactivity of adjacent TA lengths (e.g., TA6 and TA7).
- Even with optimization, there is a level of risk for amplification bias or slippage from RPA in an AT-rich low complexity region.
- Assumptions are made that there are reference standard samples and partner access to labs, but these have not been secured.
- Results are analytical/preliminary-clinical, not enough to meet clinical-grade diagnostic performance or to be validated for regulatory purposes.

Conclusion

The same genetic variation that causes Gilbert syndrome is also responsible for determining the safety of the patient receiving irinotecan or atazanavir, and as of now, the lab testing a sample to confirm the variant requires a sequencer or thermal cycler. It is really an attempt to bridge that gap. If the crRNA engineering methods already proven to be effective with single-nucleotide mutations and pathogen DNA can be pushed to the limit, can they distinguish between alleles that vary by one TA repeat? If it is as good as sequencing, or even better, in the pilot testing, then it would indicate that there is a cheaper and faster alternative to getting this result, not in a specialized lab, but in some sort of general lab. If it does just as well, or even better than sequencing, in the pilot testing, then there is a cheaper and faster alternative to getting this result, not in a specialized lab, but in some sort of general lab, and this could be repeated with other repeat-based genetic markers in the future.

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